Oline A. Ranum

AI Engineer | Applied Machine Learning | Physicist

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AI engineer with 10 years of study and research in data analysis, modeling and engineering, including internships at **CERN** and **Lawrence Berkeley National Laboratory**. Expertise spans computer vision, natural language processing, multimodal and low-resource learning, data collection and evaluation frameworks. Recognition includes **thesis prizes**, **ELLIS Honours**, a **cum laude** distinction and **peer-reviewed publications**. **Founded and led an AI start-up**: assessed market needs, worked on early technology and data acquisition, managed projects and secured a **€100k grant**.

Experience

PhD Researcher & Research Assistant, Cognitive Vision Lab, University of Surrey, UK — 2025 – present

- Develop modeling, evaluation and explainability methods for low-resource, multimodal AI in sign language.
- Research assistant on SignGPT (Surrey, Oxford, UCL): data capture, hardware and research software.
- Communicate model behavior and limitations to technical and non-technical audiences.

Data Annotator & Quality Controller, Accenture, Amsterdam — 2025

- Quality control of LLM and VLM training data: annotation, consistency and model limitations.

Researcher, SignLab Amsterdam, Netherlands — 2024

- Built automatic annotation methods and graphical interfaces for linguistic research.

CEO & Founder, DeepSign AS, Oslo — 2021 – 2022

- Led an AI start-up developing machine translation for Norwegian Sign Language.
- Owned data collection, technical development, finances, partnerships and team coordination.
- **€100k STUD-ENT grant**; ACE Cohort 2022, Demonstrator Lab; Titan.uio.no, Argument.no.

CTO & Founder, BliFlink, Oslo — 2020

- Built and managed an online tutoring platform providing free educational resources during COVID-19.
- Responsible for management, implementation and product development.

Technical Student & Research Intern, CERN, AEgIS Group, Geneva — 2019 – 2020

- Wrote software for positronium-production analysis and detector calibration for an antimatter beam.

Research Intern, Lawrence Berkeley National Lab & Berkeley Astrophysical Institute, USA — 2019

- *Nuclear physics* (Bernstein Group): detector calibration and analysis of yttrium-86 production.
- *Astrophysics*: classification of white dwarf spectra.

Research Assistant & Lab Instructor, University of Oslo, Norway — 2018

- Taught undergraduate physics labs; built a motion capture system for chaotic motion.

Technical Skills

Applied ML	Research	Engineering	Infrastructure
Computer vision	Evaluation & benchmarking	Python, C++	Linux, Git
NLP & machine translation	Explainability	PyTorch, Lightning	HPC, SLURM
Multimodal learning	Data collection & quality	Hugging Face	Docker, Apptainer
Low-resource learning	Problem framing	vLLM, Ollama	Weights & Biases, Hydra
LLMs & agentic AI		SQL, pandas	

Education

MSc, Artificial Intelligence, University of Amsterdam — 2022 – 2024

- **Cum laude** and ELLIS Honours. Geometric deep learning, computer vision, natural language processing.
- Thesis: *3D Awareness, Geometry, and Linguistics in Neural Sign Language Representations*; awarded the FNWI Master's Thesis Prize and the Amsterdam AI Thesis Award.
- Projects: SE(3)-equivariant neural rendering, steerable diffusion for scene generation.

MSc, Astroparticle and Gravitational Physics, University of Amsterdam 2020 – 2021

- Partial completion | 54 EC. Entrepreneurship, detector engineering, mathematics for physics.

BSc, Physics and Astronomy, University of Oslo 2016 – 2020

- 210 EC. Computational physics, mathematics, programming and experimental data analysis.

BSc, International Relations, University of Oslo 2015 – 2016

- Partial completion | 60 EC. International law, human rights, political theory, philosophy.

Study abroad, University of California, Berkeley, USA | 30 EC. Data-driven astronomy, nuclear engineering 2019

Summer school, Stellenbosch University, South Africa | 5 EC. *Advanced Nuclear Reactions and Applications in Astrophysics* 2019

Recognition

- **FNWI Master's Thesis Prize**, University of Amsterdam, 2025
- **Amsterdam AI Thesis Award**, best AI thesis in Amsterdam, 2024
- **ELLIS Honours**, European Laboratory for Learning and Intelligent Systems, 2023
- **STUD-ENT Grant**, Research Council of Norway, €100k, 2021

Publications

Data collection

[1] **Ranum, O.**, Wessels, D., Otterspeer, G., Bekkers, E., Roelofsen, F., Andersen, J. (2024). *The NGT200 Dataset: Geometric Multi-View Isolated Sign Recognition.* ICML 2024 Workshop on Geometry-grounded Representation Learning and Generative Modeling. openreview.net/forum?id=idkNzTC67X

[2] **Ranum, O.**, Otterspeer, G., Andersen, J., Roelofsen, F. (2024). *3D-LEX v1.0: 3D Lexicons for American Sign Language and Sign Language of the Netherlands.* LREC-COLING 2024 Workshop on the Representation and Processing of Sign Languages. aclanthology.org/2024.signlang-1.33

Field overviews

[3] Mercanoglu Sincan, O., Pelykh, A.*, Fish, E.*, Walsh, H.*, Low, J.*, Sahin, K.*, **Ranum, O.***, Asasi, S.*, Emery, S.*, Bowden, R. (2027). *Machine Translation for Sign Languages.* Annual Review of Linguistics, Vol. 13 *(accepted).* *Equal contribution.

[4] Brown, M., **Ranum, O.**, Fish, E., Proctor, H., Woll, B., Bowden, R., Cormier, K. (2026). *SignGPT and the Visual Language Toolkit.* LREC 2026 Workshop on the Representation and Processing of Sign Languages. sign-lang.uni-hamburg.de/lrec/pub/26026.pdf

Evaluation frameworks

[5] **Ranum, O.**, Fish, E., Hadfield, S., Bowden, R. (2026). *Beyond BLEU: A Case for Redefining Sign Language Translation Benchmarks.* arXiv preprint. arxiv.org/abs/2609.03734

[6] Cory, O., Ivashechkin, M., Sahin, K., **Ranum, O.**, Low, J., Fish, E., Pelykh, A., Mercanoglu Sincan, O., Bowden, R. (2026). *BackTranslation2.0: A Linguistically Motivated Metric to Assess Sign Language Production.* ECCV 2026. arxiv.org/abs/2606.28673

Modeling & methods

[7] **Ranum, O.**, Hadfield, S., Bowden, R. (2026). *What's the Point? Spatial Grammar & Index Resolution for Sign Language Recognition.* arXiv preprint. arxiv.org/abs/2606.08056

[8] Rienäcker, B., Brusa, R. S., Caravita, R., Mariazzi, S., Penasa, L., Pino, F., **Ranum, O.**, Nebbia, G. (2022). *A fiber detector to monitor ortho-Ps formation and decay.* Nuclear Instruments and Methods in Physics Research A, 1027, 166275. doi.org/10.1016/j.nima.2021.166275

[9] Hu, A., **Ranum, O.**, Pozrikidou, C., Zhou, M. (2023). *Reproducibility study of "Joint Multisided Exposure Fairness for Recommendation".* ML Reproducibility Challenge 2022. openreview.net/forum?id=A0Sjs3IJWb-

Full publication list: Google Scholar | olineranum.github.io/publications

Languages

Norwegian (native) | **English** (fluent) | Dutch (beginner)
Norwegian Sign Language (basic working knowledge) | British Sign Language (Level 1)

Talks, Supervision & Engagement

Talks	Invited talk on student entrepreneurship and innovation for social good, University of Oslo (2021); positronium detection at the AEḡIS Collaboration Meeting, CERN, and INTPART, Wits University (2019)
Supervision	Bachelor's thesis in Business Development and Innovation, VU Amsterdam (2020)
Leadership	Student representative, Physics Student Council, Board of Education and Institute Board, UiO (2017–18); founder, physics discussion groups, UiO (2018)
Outreach	Event manager, IAESTE Career Fair, Norway (2017–18); representative, SAIH (Students and Academics International Help Foundation) (2015)

References available on request.